spotter

Full Stack Developer

Weather-Aware Truck Routing

Build a **React + Django** web app that helps a truck driver choose the safest route based on **weather, load weight, and travel time**.

Requirements

The user enters:

- Origin & destination
- Departure date/time
- Load weight

The app should:

- Generate **3 route options** using a routing API.
- Sample weather every **10 / 25 / 50 miles**, using the truck's **estimated arrival time** at each checkpoint.
- Calculate weather risk using the provided risk/load rules.
- Recommend the best route, balancing **weather risk and travel time**.
- Show all routes, checkpoints, risk levels, ETAs, and route summaries on a map.
- Show a **weather heatmap along the trip corridor** with a **0–48 hour forecast slider**.

You may use third-party APIs for routing, geocoding, and weather. You do not need to build a routing engine.

Risk Rules

Use these thresholds to classify weather at each checkpoint:

Condition	Low	Moderate	High	Severe	No Travel
Wind	<25 mph	25–34	35–44	45–54	≥55 mph
Rain	<0.10 in/hr	0.10–0.25	0.25–0.50	0.50–1.00	>1.00
Snow	<0.5 in/hr	0.5–1.0	1.0–2.0	2.0–3.0	>3.0

Load rules:

- ≥55 mph → **No Travel** for any load
- 45–54 mph + >30,000 lb → **No Travel**
- 35–44 mph + >40,000 lb → **Severe**

Route Recommendation

Prefer:

1. Fewest Severe miles
2. Fewest High miles
3. Lowest average risk
4. Shortest travel times

Deliverables

- GitHub repository
- Hosted application
- **5–10 min Loom walkthrough**

Evaluation

40% Routes, Weather, risk & recommendation logic
40% UI, map & heatmap smoothness and performance
15% Code quality
5% Documentation & walkthrough

Prioritize correctness and clarity over extra features.